Lecture 2: The value of life

Economics of Health and Health Care
Spring 2026 S2-T3

Pedro Pita Barros

Outline

What is health? How to measure it?

The economic value of life

- Marginal vs total effects
- Risk premium
- Willigness to pay
- Human capital

Value of longevity

WHO definition

- How to value health? We start by looking at the value of life
- State of perfect health, and then look at variations of that state.

Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.

Discussion about alternatives

"Most criticism of the WHO definition concerns the absoluteness of the word **"complete"** in relation to wellbeing.

The first problem is that it unintentionally contributes to the medicalisation of society.

The requirement for complete health "would leave most of us unhealthy most of the time." Huber et al BMJ 2011

“it declares people with chronic diseases and disabilities definitively ill. It minimises the role of the human capacity to cope autonomously with life’s ever changing physical, emotional, and social challenges and to function with fulfilment and a feeling of wellbeing with a chronic disease or disability.”

“we propose the formulation of **health as the ability to adapt and to self manage**”

Different definition of health – what implications result?

New concept of health

- Health, as the ability to adapt and to self-manage – still, keeping the physical, mental and social domains.
- Can it be useful?
  - Doctors can tell patients how to adjust their lifestyle instead of treating symptoms with a drug (for chronic conditions)
  - Acute conditions do not change
  - Still needs operational conceptualization for measurement purposes
  - There are several measures in the literature for health but most do not capture capacity to cope and to adapt

Value of life:

From a physician's perspective:

"**health and life have infinite value**, is the usual saying, although it is perhaps hard to take it in all its implications (...) in economics, **the value of a good is the replacement cost.** In health, own life and health have an infinite value in one's own view. But **in a population view, the substitution cost of a life is virtually zero**" [own translation]

Alexandre de Sousa Pinto, Forum de Economia da Saúde.

Value of health vs. value of life

A distinction to be made: Marginal improvements in health vs. Life-threatning effects

For the first, the economists' approach seems to be not very far from reality, even if people do not realize it

For the second, we usually engage in ethical discussions.

- Economic analysis has found a way to overcome the ethical issue of giving value to a life.
- It adopted the concept of ***value of a statistical life*.**
- Instead of evaluating the value of a life of a specific person, one evaluates in an abstract way, the value of saving one life in the population (without knowing which life)

How to compute the value of life?

There are several approaches to compute the value of a statistical life

Most common

Risk premium – revealed preference

Willingness to pay – stated preference

Used in the past

Human capital (measured by income profile) – mostly abandoned for economic analysis, still used by courts

Risk Premium

What is the wage premium an individual is willing to accept to compensate for a job with a higher health risk?

Using this value and further information on the risk characteristics of the job, one can infer the value implicitly given to a statistical life.

Example: risk premium approach

- Two identical jobs, two identical workers
- Construction workers, risk of death at work
- In job A, risk is 2 per 10,000 per year
- In job B, risk is 1 per 10,000 per year
- Job A pays 500€ more per year to compensate for risk
- In 10,000 it “pays” for one extra life lost – value of a statistical life = 10,000x500€ = 5M€

Risk premium

- Moore e Viscusi (1990) take this idea to the data
- Change in risk => change in wage (for a reference group of people)
- Regression of wages of different jobs on
  - Individual characteristics - age, gender, education, experience, etc…
  - Characteristics of the position (like the number of non-fatal accidents per 1000 workers, the number of fatal accidentes per 1000 workers, etc…)

Risk premium approach - regression

Log wage as a function of				
Experience		0.028		
Experience^2		-0.0006		
Sex			-0.288	
Education		0.044		
poor health status		-0.079		
Blue Collar		-0.064		
Union status		0.182		
Death risk		0.00747		
Weighted replacement rate	-0.00805			
VSL				4.3M USD

Risk premium

The usual effects for the coefficients of the wage equation

The more interesting ones are the last two - related to risk of death

The first is the effect associated with an increase in death risk directly

The second is the compensation received in case of accident and permanent incapacity to work – compensation in case of an adverse event to health

Risk premium

- The coefficient of the death risk variable gives how much the worker requires as wage premium for an increase in the probability of death
- Example: from 3 to 4 deaths / 100 000 workers, what happens?
- If the death risk increases 1 per 100 000, the (log) of wage should increase in 0.00747.
- Effect of replacemente rate - the wage decreases 0.00805 multiplied by the replacement rate value
- Its average value is 0.544

Risk premium

The combined effect means that the total risk premium is

$$0.0030908 = 0.00747 - 0.00805 \times 0.544$$

The change in the log of wage is approximately the rate of wage increase demanded for higher risk = 0,3%

Risk premium

- Multiplying by the value of the average hourly wage, 7,01 USD, gives 0,02167 USD (per hour)
- Take a year with 2,000 hours of work, it means 43,33 USD per worker = 2 000x0,01267
- to save on average a life in 100 000 workers => the total amount of compensation needed is
- 100 000 X 43,33 = 4,3 million dollars (1981 prices) (at 2018 values, about € 8M)

Willingness to pay

Take a risk reduction in the number of premature deaths in a given year of 1 per 100 000 individuals

If each person is willing to pay A to have such risk reduction, per each death avoided the amount collected would be, aggregating, A x 100 000.

Willingness to pay

$$p_0 U(X, H_0) + (1 - p_0) \times U(X, H = 0) =$$
$$= p_1 U(X - A, H_1) + (1 - p_1) U(X - A, H = 0)$$

- Take a first-order approximation to the right-hand side and rearrange:

$$\frac{U(X, H_0) - U(X, H = 0)}{p_1 U'(X, H_0) + (1 - p_1) U'(X, 0)} = \frac{A}{\Delta p}$$

- The value of a statistical life is given by: $VSL = \dfrac{A}{\Delta p}$

Estimates in the literature

- Johannesson, Johansson and O'Connor (1996) - estimate the value of a statistical life as between 8,9 million USD and 7,4 million USD
- Viscusi and Aldi (2003) - survey of several studies - median of 6,6 million USD
- Based on seatbelt usage, Hakes and Viscusi (2004) - between 2,2 and 7,9 million USD

Estimates in the literature

- Blomquist (2004) - review of existing studies - values in this type of range
- European Union: €3.6 million
- Schoeters et al., 2021: €6.2 million (2021 values)
- Canada: €4.1 million

Estimates in the literature

- US Environment agency: 9,1 M€
  - From the New York Times: “The Environmental Protection Agency set the value of a life at $9.1 million last year [2010] in proposing tighter restrictions on air pollution”
- US Food and Drug Admin:
  - The Food and Drug Administration declared that life was worth $7.9 million (2016), up from $5 million in 2008, in proposing warning labels on cigarette packages featuring images of cancer victims.
- US transportation: 9.6M USD (2016) (about €8.13 Million)

Estimates in the literature

- Viscusi as mentioned in NYT (2011): 8.7M USD
- Viscusi, Huber and Bell (2024): 10.85 USD
- Kniesner and Viscusi (2019): 10M USD (2017) = 12.88M USD (2025) = 12.36M€ (updated according to inflation and exchange rate)
- Ananthapavan et al (2017), Australia = 5.6 M€

Australian Government
Department of the Prime Minister and Cabinet
Office of Impact Analysis

OIA
The Office of Impact Analysis

Value of statistical life

January 2026

Key points

- Willingness to pay is an appropriate way to estimate the value of reductions in the risk of unforeseen fatality by chance – known as the value of statistical life.
- Based on international and Australian research a credible estimate of the value of statistical life is $5.9 million and the value of statistical life year is $253,000 in 2025 dollars.
- These estimates are based on a number of assumptions. A sensitivity analysis should be undertaken as part of the cost-benefit analysis.

- 5.9 Million Australian Dollars = 3.48 M €

Human capital

- Sees the value of life as the present value of all future income streams of the individual
  - Often used by the courts
  - The concept of statistical life is not required
- It has severe limitations
  - Value of non-paid activities or withour a formal market
  - Future income is not the same thing as willingness to pay
  - How to value interventions with unknown beneficiaries?

Relationship of VSL and VOLY

- Another relationship is:

$$VSL = VOLY \sum_{t=0}^{T} \frac{P_t}{(1+r)^t}$$

- Where P_t is the probability of reaching period t and r is the discount rate. If no discounting on value of life, assuming that the person reaches life expectancy then dies (extreme assumptions),

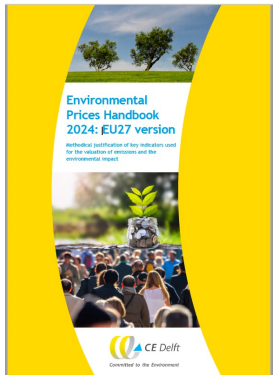
$$VOLY = \frac{VLS}{Life\ expectancy}$$

- VSL: monetary value to an instant death
- VOLY: value attached to one additional year of life expectancy
- Next step: weight value of one additional year of life by quality of life
- Discussion in practice: use VSL or estimate directly VOLY?

EU28 VOLY = €70 000 (2016)

Netherlands VOLY: €85 000 (2024)

UK: 40 000 to 60 000 GBP (2020) (45000€ to 69000€)



Valuing longevity

- Another approach is to value increases in life years
- Becker, Philipson and Soares (2005) American Economic Review – measure value of extra life according to value of consumption gained

$$V(Y,S) = \max_{\{c(t)\}} \int_0^\infty \exp(-rt)S(t)u(c(t))dt$$

$$s.a. \int_0^\infty \exp(-rt)S(t)y(t)dt = \int_0^\infty \exp(-rt)S(t)c(t)dt$$

(*S* is survival function)

Value of longevity

- Value *W* of added survival *S'*:

$$V(Y',S') = V(Y'+W,S)$$

- Some simplifying assumptions (representative agent, smooth consumption over time):

$$V(Y,S') = u(y) \times A(S), A(S) = \int_0^\infty \exp(-rt)S(t)dt$$

What to take from this?

- Need to operationalize concept of health with outcome measures relevant to inform policy
- Unsolved challenge: accounting for individuals' capacity to adapt
- Valuing life is needed to inform decisions about resource allocation to different options
- Avoid (some) ethical dilemmas by using the concept of value of statistical life
- Use adequate techniques to compute it, though many complexities are not yet fully addressed (value of life at the end-of-life vs at your 20s)
- It can be done

Readings

- Huber et al. (2011) How should we define health?, British Medical Journal, 343 (available online)
- PP Barros, chap. 3
- Folland, Goodman and Stano, Economics of Health and Health care, chap. 4 (a couple of pages).